International Journal of Innovation and Technology Management
Vol. 15, No. 6 (2018) 1950005 (35 pages)

DOI: 10.1142/S0219877019500056

Boundary Spanning, Group Heterogeneity and Engineering Project Performance

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Received 4 May 2017
Accepted 5 November 2017
Published 4 July 2018

This study examines the concept of intra-organizational links as a way for boundary spanners to bring into the project group the information needed to deal with task uncertainty. Several studies have shown that heterogeneous groups are superior to homogeneous groups when novel or creative solutions need to be developed to deal with tasks characterized by high task uncertainty. For boundary spanners in engineering project groups, it is proposed that cross-departmental technical advice links are another source of the information needed to deal with task uncertainty. An empirical test supports the proposition that for high-performing project groups, boundary-spanning technical advice links may compensate for a lack of internal group heterogeneity and vice versa. This is not the case for low-performing project groups. Implications of these finding are presented, including the direction that the open innovation research stream might take to address the findings of this study.

Keywords: Boundary-spanning; group heterogeneity; task uncertainty; link multiplexity; open innovation; project group performance.

1. Introduction

Open innovation has become a popular organizing framework in many R&D/innovation/technology management journals since 2003 when Chesbrough published his first book on this subject. Chesbrough argues that to be innovative firms need to supplement their internal R&D activities with information from external entities such as customers, suppliers, and universities. Since the publication of Chesbrough book on open innovation, articles in R&D/innovation/technology management journals has mostly focussed on the boundary that separates the firm from its environment. For example, recent articles have focussed on external search strategies [Sofka and Grimpe (2010); Kohler *et al.* (2012)], different types and forms of openness [Dahlander and Gann (2010)], development of infrastructure to support

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open innovation [Minshall *et al.* (2014)], measuring the degree of openness [Michelino *et al.* (2015)], mechanisms for developing processes to support open innovation [Wan and Quan (2014); Ebersberger *et al.* (2012)], the influence of R&D investment on open innovation [Berchicci (2013)], developing the capacity to assimilate external information [Zobel (2017)], and alliance relationships [Wang *et al.* (2017)]. At the same time, there has been a dearth of recent studies in these R&D/innovation/technology management journals on intra-organizational links as a way for boundary spanners to bring into the project group the information needed to deal with task uncertainty.

There is evidence to suggest that the boundary that divides the firm from its external environment may not be the critical demarcation for successful innovation. For example, Miller *et al.* [2007] in a large sample study of R&D groups within incumbent firms found that most innovative ideas come from within the firm and from another division rather than from outside the firm. In a study of banking innovation, Olivera and von Hippel [2011] found many of the ideas for new services came from customers, but this percentage varied considerably by type of innovation, and joint- and firm-sourced ideas were still very important. In addition, most of these ideas were incremental improvements on platforms that already existed, not breakthrough innovations that were new to the industry.

At the juncture of the two sectors (manufacturing and services) and in a study of nine significant, new service innovations by incumbent manufacturing firms, all but one originated within the firm [Ettlie and Rosenthal (2012)]. The absence of external influence on the innovation process in these three published empirical studies suggests that the rush to assume that the boundary external to the firm is the essential focus of the innovation process may be a misplaced positivist approach to this subject, and a correction to this bias is needed. Although it is true that many examples of significant customer, supplier, competitor and crowd-sourced innovations have been documented, there is no extant evidence that these are the only or the dominant forms of successful innovation processes [West and Gallagher (2006)].

Consequently, it may be that there are alternative ways of boundary management within, especially the large, incumbent firm. In contrast to studies of open innovation, the focus of this paper is on boundary spanning between a focal project group and other project groups within the same organization, thereby excluding contacts with entities external to the focal organization. Our aim is to show how important boundary spanning activities *internal* to an organization are for technology-intensive firms.

With the increased adoption of team-based project groups, organizational researchers have come to realize that team performance is not merely an outcome of the internal function of teams but is also the outcome of team-based boundary-spanning activities [Ancona and Caldwell (1992a,b); Joshi (2006); Joshi *et al.* (2009); Reagans *et al.* (2004)]. However, only a few studies have focussed on the antecedents of boundary-spanning activity within a team or project group [e.g. Marrone *et al.* (2007); Faraj and Yan (2009)]. For example, Drach-Zahavy and Somech [2010,

p. 144] note that boundary-spanning studies: "...tended to examine the consequences of boundary-spanning and paid less attention to the antecedents that constrain or enhance the extent to which teams engage in boundary activities."

Antecedents of boundary-spanning that have received the attention of researchers include the effects of group heterogeneity on team innovation [Somech and Khalaili (2014)]. Some researchers have focussed on why high-performing project groups in an organization are able to effectively deal with different degrees of task-related uncertainty whereas low-performing groups are not. For example, Tushman [1979] argued that high-performing project groups are able to match their ability to find and process the information needed to successfully deal with the task-related problems facing the project group, whereas low-performing groups are not able to do so. Scholars who have adopted a team or project group perspective have traditionally addressed this problem by analyzing variables such as the amount of group heterogeneity and the climate of the group (e.g. whether differences are tolerated) and how these group characteristics influence the group's information-processing capabilities. For example, heterogeneous groups may be more successful than homogeneous groups when it comes to solving difficult task-related problems.

The type of information needed when tasks are perceived by group members as easy to analyze (or analyzable) is very different from the type of information needed when tasks are perceived as difficult to analyze (or unanalyzable): difficult-to-analyze problems require a more diverse information base. The concept of "strength of tie" [Granovetter (1973)] is one way of characterizing the qualitative differences in the information received by group members from organizational sources outside the group. Therefore, in this study, strength of tie (i.e. weak technical advice links) will be used as a way to explicitly bring into the analysis a consideration of the diversity of the information contained in a technical advice link that goes to organizational members outside the group.

The question then arises whether it is the characteristics of the group (i.e. the heterogeneity of the individual characteristics of group members) or the use of boundary-spanning technical advice links that enables high-performing groups to effectively deal with difficult-to-analyze task-related problems. In this paper, it will be argued that the relationship between the degree of task uncertainty facing a project group and a group's use of boundary-spanning technical advice links may be contingent upon the internal characteristics of each project group. That is, group heterogeneity will exert a moderating influence on the extent to which boundary-spanning technical advice communication links are used to gather the information needed for a project group to deal with difficult-to-analyze task-related problems. For example, homogeneous groups that may lack the internal diversity needed to successfully deal with difficult-to-analyze problems may be able to compensate by using boundary-spanning technical advice links to bring into the group new information to increase the diversity of the group's information base, whereas the diversity of information that may be available in a heterogeneous group may make it unnecessary to go outside that group for technical advice.

2. The Role of Informational Diversity

2.1. *Task analyzability and informational diversity*

In their review of the literature, Baker *et al.* [1980, pp. 40–41] argued that the use of diverse information enhances problem-solving by increasing the range of possible solutions that will be recognized and evaluated. Baker *et al.* [1980] pointed out that it is the quality of the diverse information, more than its quantity, which enhances idea generation. Mannix and Neale [2005] argue that a diverse group is especially important when a group needs to deal with complex tasks or when a group needs to develop a solution to a challenging task-related problem. In contrast, simple, routine tasks require a group to follow a well-defined course of action, so access to diverse information is typically not needed. The diversity of the information received is important since the development of novel or unique solutions to difficult-to-analyze problems may require finding and using information that might normally not be considered [Fischer (1980)] or that is, at first glance, only peripherally related to the information normally received [Menzel (1962)]. Diversity is introduced into the information base by contacts with diverse colleagues (in terms of skills, area of specialty, and backgrounds), both inside and outside a group or organizational unit [Kasperson (1978); Pelz and Andrews (1976); Dokko *et al.* (2014); Ely and Thomas (2001)]. Innovative ideas often occur from the recombination of diverse knowledge [Henderson and Clark (1990); van Knippenberg *et al.* (2004)].

Diverse information may come from sources outside the project group. Johnston and Gibbons [1975, p. 32] noted that a decision maker usually tries to rely on the knowledge already possessed, but when this knowledge is insufficient will seek information from other sources. However, it is the "information with which the problem-solver is least familiar and which is least readily accessible to him at the commencement of the innovation that in practice provides the greatest contribution to solving the technical problems." Johnston and Gibbons recommend the development of communication links outside the project group in order to tap sources of new and diverse information. In short, crossing organizational boundaries may allow members of a group to access diverse knowledge in other parts of the organization [Burt (2004); Fleming *et al.* (2007); Perry-Smith and Shalley (2003); Sosa (2011)].

2.2. *Strength of tie and informational diversity*

The type of information needed when tasks are perceived by group members as easy to analyze (or analyzable) is very different from the type of information needed when tasks are perceived as difficult to analyze (or unanalyzable): difficult-to-analyze problems require a more diverse information base. The concept of "strength of tie" [Granovetter (1973)] is one way of characterizing the qualitative differences in the information received by group members from organizational sources outside the group. Therefore, in this study, strength of tie (i.e. weak technical advice links to individuals outside the group) will be used as a way to explicitly bring into the analysis a consideration of the diversity of the information contained in a technical advice link that goes to organizational members outside the group.

If the information needed to solve difficult task-related problems is not readily available from co-workers within the project group, technical advice may be sought from individuals outside the group. Granovetter [1973, p. 1361] has proposed that ties or communication links between individuals have varying strength, ranging from strong to weak. Weak ties are linkages between people that, although infrequent, connect individuals with access to different information sources. What is unique about weak ties is that they serve as bridges between groups or clusters of strong ties. "It follows that individuals with few weak ties will be deprived of information from distant parts of the social system and will be confined to the provincial news and views of their close friends" [Granovetter, 1982, p. 106]. According to Granovetter, strong ties are more important than weak ties in promoting information flows within a social system such as a group, whereas weak ties are more important in promoting information flows from outside that group or social system. The advantage of weak ties is that they may bring into a project group new, diverse information that is not available within the group. In short, weak ties serve as information bridges across cliques of strong ties and thus can provide access to new information not available in their strong-tie relationships [Burt (2004); Perry-Smith (2006); Dokko *et al.* (2014); Fletcher (1998)].

Wellman (1981, p. 186) also believed that weak ties are an important mechanism by which new information enters a group or organizational unit. "Strong ties link network members to persons of similar backgrounds who travel in the same social circle as they do, and the more ramified weak ties link them to other, dissimilar social circles." That is, weak ties are likely a source of new and diverse information.

Hansen [1999] found that weak inter-unit ties help a project group search for useful information that may be available in other organizational units, but such weak ties may make it difficult to transfer knowledge that is tacit or ambiguous. Weak ties are best used to search for new information and to transfer information that is directly applicable to the group seeking such information. Related arguments have been proposed by Moran [2005], Oh *et al.* [2006], and Reagans and McEvily [2003].

When project members do not have all the appropriate information within their group to deal with a difficult-to-analyze task-related problem, they can use technical advice links to seek information and ideas from outside the formal work group [Allen (1977)]. Boundary-spanning activity is likely to increase when task-related problems are difficult to analyze [Ito and Peterson (1986)]. If these boundary-spanning links are also weak links, group members may acquire new and diverse information that may not be available from the strong technical advice links that often characterize most of the ties to co-workers within the group. Weak boundary-spanning technical advice links to individuals outside the group may thus provide group members the diverse information needed to deal with difficult-to-analyze task-related problems [Reagans and Zuckerman (2001)].

3. Group Characteristics

3.1. *Group heterogeneity*

One advantage of groups is that they can contain a diversity of skills, abilities, attitudes, and areas of know-how that may be lacking in any one individual [Bantel and Jackson (1989); Hoffman (1959); Hoffman and Maier (1961); Triandis *et al.* (1965)]. Groups whose members are diverse with regard to background and perspective outperform groups with homogenous members on tasks requiring problem-solving and innovation [Harrison and Klein (2007); Jackson (1992); Jackson and Joshi (2011)]. The more diverse the skills, backgrounds, and experiences of group members, the more these members should be a good source of the diverse information needed to solve difficult-to-analyze task-related problems [Jehn (1995); Jehn *et al.* (1999); Horwitz and Horwitz (2007); Wanous and Youtz (1986)]. In short, a heterogeneous group is one source of the diverse information that may be needed to solve difficult-to-analyze task-related problems.

But what can be done if a group is too homogeneous to provide the diverse information needed to deal with difficult-to-analyze problems? One possibility is to compensate for a lack of internal group heterogeneity by using weak boundary-spanning technical advice links to bring into the group new, previously unknown information. By such a process, the diversity of the information base in the group is enlarged even if internal group heterogeneity is low. Consequently, the need to go outside the project group for technical assistance in solving difficult-to-analyze task-related problems may depend upon the diversity of information that exists within the group [Pelz and Andrews (1976); Smith (1970, 1971); Zenger and Lawrence (1989)].

The above arguments suggest that a project group's use of weak boundary-spanning technical advice links may depend upon whether the problems a group faces are easy or difficult to analyze and whether the group is homogeneous or heterogeneous. An empirical *interaction* is being predicted: group heterogeneity moderates the relationship between task analyzability and the proportion of the technical advice network that consists of weak boundary-spanning links. Consequently, the effects of changes in task analyzability upon the proportion of the technical advice network that consists of weak boundary-spanning links are described below, first for the case when group heterogeneity is low and then for the case when group heterogeneity is high. In short, as task analyzability increases or decreases, the changes that occur in the proportion of a group's technical advice network that consists of weak boundary-spanning links may be moderated by the heterogeneity of the group.

3.1.1. *Low group heterogeneity*

When the task-related problems encountered by a project group are *easy* to analyze, diverse information is not really required so there is little need for weak boundary-spanning technical advice links. For easy, analyzable problems, co-workers within the group who are probably already familiar with a problem and can readily suggest an acceptable solution should be good sources of any technical advice that may be needed [Bowers *et al.* (2000)]. Therefore, when group heterogeneity is low and task-related problems are easy to analyze, the proportion of a group's technical advice network that consists of weak boundary-spanning links should be relatively low.

If the problems encountered by a project group are *difficult* to analyze, a more diverse informational base is probably needed to successfully solve these problems. Since a homogeneous group is not likely to be a good source of the diverse information needed for solving difficult-to-analyze task-related problems [Bowers *et al.* (2000)], diverse information must be imported from outside the group by way of weak boundary-spanning technical advice links. Thus, when group heterogeneity is perceived to be low and task-related problems are difficult to analyze, the proportion of a group's technical advice network that consists of weak boundary-spanning links should be relatively high.

3.1.2. *High group heterogeneity*

When task-related problems are *easy* to analyze, there is likely little need for the diverse information that may be available from a heterogeneous group. Indeed, asking heterogeneous co-workers for technical advice on easy-to-solve problems may only increase the potential for intra-group conflict which would likely be detrimental to group functioning [Jehn (1995); Pelled *et al.* (1999); Williams and O'Reilly (1998); Zenger and Lawrence (1989); Slepian (2013)]. Consequently, little, if any, technical advice is likely to be sought from co-workers within the group, and any technical advice that is needed may be sought from individuals outside the group in order to avoid the possibility of inter-group conflict [Ancona and Caldwell (1992b)] or the problems associated with dealing with too much diversity within a group [Mors (2010)]. (As the *proportion* of the technical advice network that is within the group goes down, the *proportion* of the technical advice network that goes outside the group will go up.) Therefore, the proportion of a group's technical advice network that consists of weak boundary-spanning technical advice links should be relatively high.

As the problems encountered by a group become more *difficult* to analyze, a more diverse informational base is likely to be needed. When group heterogeneity is perceived to be high, the diverse information needed to deal with difficult-to-analyze task-related problems is likely available from co-workers within the group, so there is probably little need for group members to use weak boundary-spanning technical advice links. Therefore, the proportion of a group's technical advice network that consists of weak boundary-spanning links should be relatively low when group heterogeneity is high and task-related-problems are difficult to analyze.

3.1.3. *Summary of hypotheses*

The following hypotheses concerning the *interaction* of task analyzability with perceptions of group heterogeneity are presented, assuming the climate of the group (i.e. tolerance of differences) is held constant at its average value: Hypotheses are stated in a form (i.e. high and low levels of a variable) that corresponds to the methods for statistically analyzing interaction effects as described in Sec. 5 of this paper.

Hypothesis 1A: *Low group heterogeneity*. When members of high-performing groups perceive their group to have low levels of heterogeneity,

(1) a *low* level of task analyzability (i.e. tasks are difficult to analyze) will result in a higher proportion of the technical advice network that consists of weak links that go to individuals outside the group but within the organization;
(2) a *high* level of task analyzability (i.e. tasks are easy to analyze) will result in a low proportion of the technical advice network that consists of weak links that go to individuals outside the group but within the organization.

Hypothesis 1B: *High group heterogeneity*. When members of high-performing groups perceive their group to have high levels of heterogeneity,

(1) a *low* level of task analyzability (i.e. tasks are difficult to analyze) will result in a low proportion of the technical advice network that consists of weak links that go to individuals outside the group but within the organization;
(2) a *high* level of task analyzability (i.e. tasks are easy to analyze) will result in a higher proportion of the technical advice network that consists of weak links that go to individuals outside the group but within the organization.

Hypothesis 1C: Hypotheses 1A and 1B will *not* be supported for low-performing project groups.

Hypothesis 1A and 1B are summarized in Table 1, Matrix 1.

3.2. *Group climate: Tolerance of differences*

High group heterogeneity can be a double-edged sword. On the positive side, high group heterogeneity could increase the problem-solving potential of the group. On the negative side, high group heterogeneity may increase the difficulty of building

Table 1. Summary of hypotheses.

Matrix 1. The proportion of high-performing group's technical advice network that consists of weak boundary-spanning links (% weak technical advice links) according to Group heterogeneity and Task analyzability		
High task analyzability	H1A. Low % weak technical advice links	H1B. High % weak technical advice links
Low task analyzability	H1A. High % weak technical advice links	H1B. Low % weak technical advice links
	Low group heterogeneity	High group heterogeneity
Matrix 2. The proportion of high-performing group's technical advice network that consists of weak boundary-spanning links (% weak technical advice links) according to Group heterogeneity and Tolerance of differences		
Low tolerance of differences	H2A. Low % weak technical advice links	H2B. High % weak technical advice inks
High tolerance of differences	H2A. High % weak technical advice links	H2B. Low % weak technical advice links
	Low group heterogeneity	High group heterogeneity

interpersonal relations: there may be more interpersonal conflict and lower group cohesiveness when group heterogeneity is high [Chi *et al.* (2013); Cronin and Weingart (2007); King *et al.* (2009); Milliken and Martins (1996); Pelled *et al.* (1999); van Knippenberg and Schippers (2007); van Knippenberg *et al.* (2004); Williams and O'Reilly (1998)]. Perceived differences in knowledge, skills, and abilities may limit the degree to which group members are willing to communicate and interact with one another [Jackson *et al.* (1995)]. Whether heterogeneous co-workers are perceived by group members in a positive or a negative light may partially depend upon the climate of the group.

Homan *et al.* [2007] found that informationally diverse groups perform better when in a climate that was pro-diversity compared with a climate that was pro-similarity. Pro-diversity beliefs make it much easier for heterogeneous groups to benefit from their diversity. "Pro-diversity beliefs may thus increase the likelihood that groups benefit from their diversity by inviting group members to actively solicit new information and perspectives from fellow group members and, thereby, stimulate performance" [Homan *et al.* (2007, p. 1192)]. Group performance was also found to be affected more by pro-diversity beliefs in informationally diverse groups than in informationally homogeneous groups. The performance of homogeneous groups was not greatly influenced by diverse beliefs; groups with pro-diversity beliefs did not perform much better than groups with pro-similarity beliefs. Smith *et al.* [2005] found that individuals in firms with a climate that supports risk taking are more open to new and novel information that leads to knowledge-creation capability. Somech and Drach-Zahavy [2013, p. 684] found that "group heterogeneity promotes creativity which then enhances innovation implementation but only when climate for innovation is high."

Therefore, the second group characteristic to be considered is the climate of the group [Ashkanasy *et al.* (2000); Falcione, Sussman and Herden (1987)]. Climate is defined as "the shared perceptions of the employees concerning the practices, procedures, and the kinds of behaviors that get rewarded, supported, and expected in a setting" [Schneider (1990, p. 384)]. Different groups or units within an organization may have different levels of a specific climate [Zohar and Luria (2005)]. Zohar and Luria also found that perceptions of organization climate were greatly influenced by an individual's perceptions of that individual's group-level climate.

Since the emphasis in this study is on finding and using diverse information to aid in solving difficult-to-analyze task-related problems, the appropriate climate dimension would seem to be "tolerance of differences," the degree to which different behaviors, backgrounds, ideas, or approaches to problem solving that may characterize a heterogeneous group are tolerated, accepted, or are otherwise perceived to be in accordance with group norms [Jehn (1995, p. 263)]. Chatman and Flynn [2001] found that group norms mediated the effects of heterogeneity on work processes and outcomes. Heterogeneous groups may be a good source of ideas only when group norms encourage trust and idea sharing. Otherwise, "sharing information and novel perspectives is risky because of the potential for social ostracism or dilution of individual credit" [Chatman *et al.* (1998, p. 756)]. In short, the degree to which differences are tolerated will determine, to a large degree, whether contact with heterogeneous co-workers within the group is seen as a possible source of interpersonal conflict and thus as something to be avoided, or as a possible source of diverse information and thus as something to be utilized in the problem-solving process [Hoffman *et al.* (1962); Nishii (2013); van Knippenberg *et al.* (2007); Ely and Thomas (2001)].

When tolerance of differences is low, group members may avoid seeking much technical advice from heterogeneous co-workers within the group who are perceived as being "different" since such interactions may only increase the potential for interpersonal conflict. Increasing the degree of tolerance should increase the ease with which group members can access heterogeneous co-workers for the technical assistance that may be needed for solving difficult-to-analyze task-related problems.

The above arguments suggest that the effects of an increase in tolerance of differences upon the proportion of the technical advice network that consists of weak boundary-spanning links may depend upon whether work group heterogeneity is perceived to be low or high.

3.2.1. *Low group heterogeneity*

When group heterogeneity is low, there are likely few "differences" to tolerate when it comes to interacting with co-workers within one's group. When tolerance of differences and project group heterogeneity are both perceived to be low, work group members may be reluctant to go to anyone other than co-workers within the same project group for technical advice [Bowers *et al.* (2000); Zenger and Lawrence (1989)]. Such groups are likely to exhibit an intergroup bias (Brewer, 1979) that results in "…less positive reactions to others categorized as dissimilar, or of out-group status, than to others categorized as similar or of in-group status" [van Knippenberg *et al.* (2004, p. 1014)]. Consequently, members of the project group may tend to stick to themselves and not be inclined to seek much technical advice from members of other groups. As a result, the proportion of a group's technical advice network that consists of weak boundary-spanning links may be relatively low when group heterogeneity and tolerance of differences are both perceived to be low.

When group heterogeneity is perceived to be low, but tolerance of differences is perceived to be high, it may become more acceptable to seek any technical advice that may be needed from members of other groups, even if these individuals are perceived to be dissimilar or of out-group status and not just seek technical advice from co-workers within the group. Thus, the proportion of a group's technical advice network that consists of weak boundary-spanning links may be relatively high (compared to when tolerance of differences is low) when group heterogeneity is perceived to be low but tolerance of differences is perceived to be high. However, the increase in the proportion of weak boundary-spanning technical advice links when tolerance of differences is perceived to be high is not likely to be much greater than when tolerance of differences is perceived to be low [Homan *et al.* (2007)].

3.2.2. *High group heterogeneity*

When tolerance of differences is low but project group heterogeneity is perceived to be high, asking highly heterogeneous co-workers for technical advice is likely to be avoided and could be a source of interpersonal conflict [Mannix and Neale (2005); Milliken and Martins (1966); van Knippenberg *et al.* (2004); van Knippenberg and Schippers (2007); Williams and O'Reilly (1998)]. Consequently, the number of technical advice links to co-workers within the group may be low. High group heterogeneity may make it difficult for group members to get along, so members may seek technical advice from kindred spirits outside the group [Ancona and Caldwell (1992b); Joshi (2006)]. Indeed, members of heterogeneous teams may prefer to associate with members of other teams that speak the same professional language [Somech and Khalaili (2014)]. Consequently, the proportion of a group's technical advice network that consists of weak boundary-spanning links may be relatively high when group heterogeneity is high and tolerance of differences is low.

Although a heterogeneous group may contain a diversity of perspectives and approaches to problem solving, it may not be sufficient just to have a diverse group if this diversity is not utilized by the members of the group [van Knippenberg *et al.* (2004)]. When the heterogeneity of the group is perceived to be high and tolerance of differences is also high, the potential for interpersonal conflict within the group is likely minimized, and it becomes acceptable to contact heterogeneous co-workers within the project group for any technical advice that may be needed [Hobman *et al.* (2004)]. Therefore, there may be little need to use boundary-spanning links for technical advice. Consequently, the proportion of a group's technical advice network that consists of weak boundary-spanning links may be relatively low when group heterogeneity and tolerance of differences are perceived to be high.

3.2.3. *Summary of hypotheses*

The following hypotheses are presented concerning the *interaction* of tolerance of differences with perceptions of group heterogeneity, assuming that task analyzability is held constant at its average value: Hypotheses are stated in a form (i.e. high, low levels of a variable) that corresponds to the methods for statistically analyzing interaction effects as described in Sec. 5 of this paper.

Hypothesis 2A: *Low group heterogeneity.* When members of high-performing groups perceive their group to have low levels of heterogeneity,

(1) a *low* level of tolerance of differences will result in a relatively low proportion of the technical advice network that consists of weak links that go outside the group but within the organization;
(2) a *high* level of tolerance of differences will result in a relatively high proportion of the technical advice links that consists of weak links that go outside the group but within the organization.

Hypothesis 2B: *High group heterogeneity.* When members of high-performing groups perceive their group to have high levels of heterogeneity,

(1) a *low* level of tolerance of differences will result in a high proportion of a group's technical advice network that consists of weak links that go outside the group but within the organization;

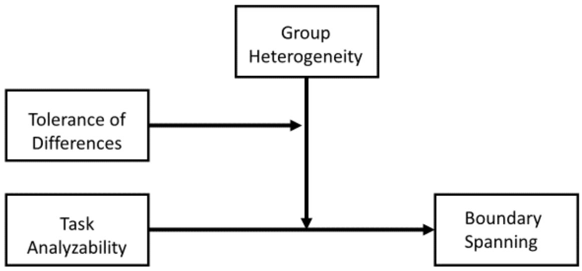

Fig. 1. Conceptual framework.

(2) a *high* level of tolerance of differences will result in a low proportion of a group's technical advice network that consists of weak links that go outside the group but within the organization.

Hypothesis 2C: Hypotheses 2A and 2B will *not* be supported for low-performing project groups.

Hypotheses 2A and 2B are summarized in Table 1, Matrix 2.

3.3. *Conceptual model*

A conceptual model for high-performing groups illustrating the relationship among the hypothesized variables is shown in Fig. 1. The extent to which group members use boundary-spanning technical advice links depends upon the level of task uncertainty (i.e. task analyzability) facing the group. This relationship is moderated by the heterogeneity of the group (e.g. when heterogeneity is high, the information needed to deal with difficult-to-analyze task-related problems may be available from co-workers within the group). Heterogeneity should increase the problem-solving potential of the group if the climate of the group is such that differences are tolerated. Consequently, the impact of heterogeneity upon the extent of boundary spanning is moderated by the extent to which differences are tolerated.

4. Methods

4.1. *Research site and participants*

The site for this research was a fairly large organization (approximately 2000 employees) whose mission is to provide system improvements and other engineering support for various complex and sophisticated in-service military weapon systems. The majority of the organization's employees are professional engineers or engineering technicians assigned to project-oriented groups. Engineers in this organization work mostly on making major improvements to existing, but highly complex systems, not on tasks related to new product development. Nonetheless, as Gales and Mansour-Cole (1995) argue, task-related problems may be perceived as difficult to analyze due to the increased awareness of constraints and sunk costs.

The employees of six of the organization's 19 departments volunteered to participate in the research study by completing a questionnaire. Each department is further divided into several project-oriented groups of around 8–10 people each. Each individual is a member of only one group. By prior agreement with the organization's top management, participants completed questionnaires at their leisure during normal work hours and returned the completed questionnaire by mail using a pre-addressed stamped envelope.

Sixteen groups with fewer than four people and groups that assigned strictly staff support or administrative duties were eliminated from the current study. Consequently, this study is based on data from 390 individuals belonging to 49 project groups, a response rate of 69%. Seventy-five percent of the respondents were men, and the average age was 38 years.

4.2. *Measures and procedures*

4.2.1. *The dependent variable — Weak boundary-spanning technical advice links*

Granovetter listed four identifying characteristics of tie strength: "The strength of a tie is a (probably linear) combination of the amount of time, the emotional intensity, the intimacy (mutual confiding), and the reciprocal services which characterize the tie" [1973, p. 1361]. However, as Krackhardt [1992] points out, tie strength has been measured many different ways. Marsden and Campbell [1984, 2012] concluded that the use of frequency of contact as a measure of tie strength tended to overestimate the strength of ties between individuals who are in the same physical work area. Frenzen and Nakamoto [1993, p. 369, italics in original] found that "frequency of contact reflects the *opportunity* rather than the motivation to transmit." Ancona and Caldwell [1992a] found that the type of external communication link (i.e. its content) was more important than frequency of external contact in predicting group performance. McEvily and Zaheer [1999, p. 1153] concluded that "infrequency of interaction may not be the best measure of weak ties" and that future research should consider other dimensions of weak ties. Consequently, link multiplexity was used in this study instead of frequency of interaction as an indicator of strength of tie. Link multiplexity has been successfully used as an indicator of tie strength [Verbrugge (1979); Kenis and Knoke (2002)]. Dahlander and McFarland [2013, p. 77] note that "...multiplex ties act much like strong ties."

Link multiplexity may be defined as the extent to which there are multiple purpose ties between organizational members [Burt (1983); Brass *et al.* (1998); Kenis and Knoke (2002); Tichy *et al.* (1979)]. An ego-centric method was used to collect the network data required to measure link multiplexity [Wasserman and Faust (1994)]. Specifically, link multiplexity was measured by asking each project group member (i.e. ego) to list the names of the people they interacted with (i.e. alters) for three purposes. The first question asked respondents to "list the people at work that you interact with to get the job done." This list of names was defined as the respondent's work network. A second question asked respondents to "list the names of the people at work that you go to for help or technical advice concerning work-related problems." This list of names was defined as the respondent's technical

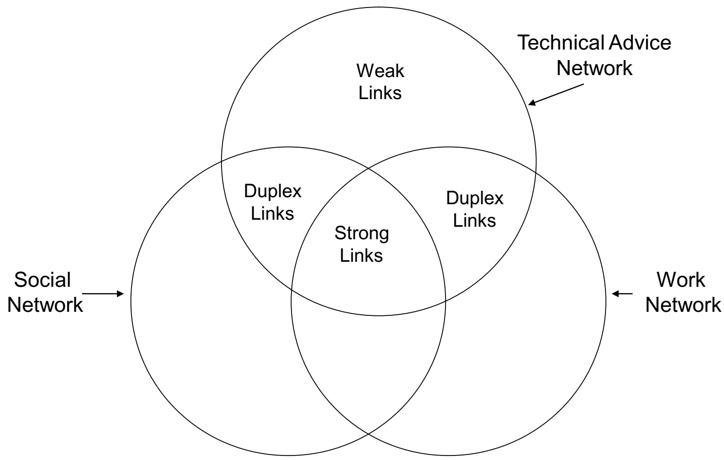

Fig. 2. Possible overlap of the technical advice network with the work and social networks.

advice network. A third question asked respondents to "list the names of the people at work that you are likely to have social conversations with." This list of names was defined as the respondent's social network. Room was provided on the questionnaire to list up to 14 names for each of these three networks. Because each respondent was given the option of listing up to 14 names per network, the number of responses varied from respondent to respondent and for each of the three networks.

As shown in Fig. 2, strength of tie is operationalized in terms of link multiplexity, the degree of overlap between a respondent's technical advice network, work network, and social network. Strong multiplex links contain all three components: work, technical advice, and social. Duplex links contain the technical advice component and either the work or social component. Weak "uniplex" links contain only the technical advice component. Consequently, these weak "uniplex" technical advice links should exhibit low "intimacy" since they exclude social connections, and many "reciprocal services" since they exclude work-related connections. Although the extent of multiplexity within a communication link was suggested by Granovetter's [1973, p. 1361] definition of tie strength, this approach to measuring the strength of tie has been utilized in only a few studies of organizational communication. Studies by Weimann [1983] and Mitchell [1987] use multiplexity as an indicator of tie strength.

The *proportion* of weak technical advice links that go outside the project group but inside the organization will be adopted as the dependent variable in this study instead of the *number* of technical advice links (see Appendix A, note 1). Such an approach is consistent with prior research [Burt (1992); Reagans and McEvily (2003); Uzzi (1996)]. Each technical advice link reported by a respondent was computer coded to indicate whether the link was inside or outside a project group as defined by published organization charts. Specifically, the proportion of a group's technical advice network that consists of weak boundary-spanning links is defined as the number of uniplex technical advice links going to individuals outside the group (as defined by the published organizational chart) but inside the organization, divided by the total number of technical advice links (inside and outside the group; weak duplex, and strong). Thus, the proportion of weak boundary-spanning technical advice links is operationalized as the proportion of the names listed in the technical advice network that do *not* appear in either of the other two networks (i.e. work and social networks) and that go to individuals *outside* the project group but *within* the organization (see Appendix A, note 2).

4.2.2. *Independent variables*

All independent variables were measured using previously published scales. Due to space limitations in the questionnaire and the time requirements for completing the network section, short scales were used whenever possible to measure the independent variables.

Perrow [1967, 1970] argued that there are two aspects or dimensions to task uncertainty. The first dimension, task variety, is related to the number of exceptions encountered in performing a work-related task. The second dimension, task analyzability, focusses on the nature of the search process that must be instituted to determine an appropriate response to an exception. The search process itself may be of two types. The first type of search process involves routine, logical analysis. Solutions are found by making "incremental adaptations from existing programs" [Perrow (1970, p. 76)]. The second type of search is unanalyzable in that ready-made solutions do not exist. Experience, intuition, or even chance may be called upon to find a novel or unique solution since programmed responses are not usually available. Perrow's inclusion of a search dimension (i.e. task analyzability) in his theoretical framework explicitly recognizes that information-processing requirements may be different for different types of tasks. Some authors [Ito and Peterson (1986); Miles (1980, p. 70)] have labeled Perrow's search dimension "task coping difficulty." Coping in response to a task exception may range from a programmed (easy) response to an unprogrammed (difficult) response.

Extending Perrow's basic argument, Daft and Macintosh [1981] argued that when tasks are difficult to analyze, qualitatively rich information is needed. The notion of qualitatively rich information suggests that Perrow's analyzability dimension is associated more with the diversity of the information processed than with the quantity of information. That is, informational *search* strategies that focus only on gathering *more* information will most likely *not* be very productive when tasks are difficult to analyze. Rather, new sources of information must be found. That is, as tasks become more difficult to analyze, the diversity of the information that must be processed should also increase accordingly [Daft and Lengel (1984, 1986); Lengel and Daft (1988); Gales *et al.* (1992)]. Since one purpose of this paper is to describe a model that includes a search dimension (i.e. the need for a diverse information base for problem solving), the emphasis here will be on task analyzability and not Perrow's other dimensions of task variety.

Questions for Perrow's dimension of task analyzability (or task coping difficulty) have been developed by several researchers [Withey *et al.* (1983) for a review]. The three questions used in the present study (Cronbach's alpha = 0.70) are adapted from Van de Ven and Ferry [1980]. Although some authors have used demographic diversity as a measure of group heterogeneity, Lawrence [1997] and Cummings [2004] have argued that in many organizational settings demographic diversity based on variables such as race, gender, or firm tenure does not directly influence network structure. Consequently, in this study group heterogeneity was measured by a three-item scale (Cronbach's alpha = 0.76) from the Michigan Organizational Assessment Questionnaire [Cammann *et al.* (1983, p. 101)]. Tolerance of differences was measured by a five-question scale (Cronbach's alpha = 0.87) adapted from Siegel and Kaemmerer [1978]. Appendix B gives the complete scales for task analyzability, group heterogeneity, and tolerance of differences.

Since the project group is the unit of analysis in the present study, all variables were aggregated to group means. One-way analysis of variance (ANOVA) was conducted with task analyzability, heterogeneity, and tolerance of differences as dependent variables and group membership as the independent variable. Using the guidelines found in Kenny and La Voie [1985], it was determined that aggregation to the group level was justified.

4.3. *Group performance*

Tushman and Nadler [1978] have proposed an information-processing model that is concerned with the information-processing abilities of project groups within a broader organizational context. In the Tushman and Nadler model, project effectiveness is contingent on matching communication-processing ability (i.e. the amount and direction of communication links) to such task characteristics as the routineness of the work (i.e. information-processing requirements). The unit of analysis in these studies was the project group. This study uses the same methodology and approach as Tushman [1979].

Tushman [1979] argued that high-performing groups are able to match their information-processing requirements to their information-processing capability, whereas low-performing groups are not able to make such a match. Consequently, high- and low-performing groups were identified and analyzed separately by Tushman. The same approach is followed here. Each project group's performance was independently evaluated by a panel of 30 top-level managers and department heads using the same questionnaire and procedures described by Tushman [1979, Appendix 3]. Based on these top-management ratings, the 49 project groups were divided into two categories, low performers ($n = 24$) and high performers ($n = 25$), by splitting the performance ratings as near as possible at the median. See Appendix A, note 3.

5. Statistical Analysis

Based on the hypotheses developed earlier, a regression equation was developed in an effort to explain variations in the proportion of a project group's technical advice network that consists of weak boundary-spanning technical advice links (i.e. uniplex technical advice links that go outside the project group but inside the organization).

The independent variables included in the regression equation were: task analyzability (Task), group heterogeneity (Het), tolerance of differences (Tol), and the hypothesized interaction terms involving these variables.

The regression equation when the dependent variable is the proportion of a group's technical advice network that consists of weak boundary-spanning technical advice links is:

$$\begin{aligned} &Y(\%\ \text{Weak boundary} - \text{spanning technical advice links})\\ &\quad = A + B1\ \text{Het} + B2\ \text{Tol} + B3\ \text{Task} + B4\ \text{Het}^{*}\text{Task} + B5\ \text{Het}^{*}\text{Tol}. \end{aligned}$$

The regression analysis was run twice, once using only the data for low-performing groups and again using only the data for high-performing groups.

Some statisticians recommend that a non-linear transformation such as the arcsine be made when proportions are used as a quantitative scale [Cohen and Cohen (1983)]. For this reason, the dependent variable (the proportion of a project group's technical advice network that consists of weak boundary-spanning links) was transformed using the arcsine. A comparison was then made of the results of the regression analysis using both the transformed and untransformed proportions as dependent variables. A careful comparison of the two sets of regressions indicated that the results were almost identical. The same results were found when a log transformation was performed on the dependent variable. Consequently, a transformed scale was not used; the regression equations reported here use only untransformed scales. For these groups, the dependent variable ranged from a minimum of 0.0 to a maximum of 0.36 with 0.15 being the mean value.

Tables 2 and 3 shows the correlations among the independent variables for low- and high-performing groups. The regression results are shown in Table 4 for low-performing groups and in Table 5 for high-performing groups. As shown in Table 4, the hypothesized interaction term between heterogeneity and task analyzability for low-performing groups is not statistically significant, indicating that these low-performing groups do not adjust the proportion of their technical advice network

Table 2. Correlations matrix for low-performing groups.

	Mean	SD	1	2	3	4
1. Group heterogeneity	5.69	0.326				
2. Tolerance of differences	4.28	0.452	−0.32			
3. Task analyzability	4.14	0.726	0.01	−0.05		
4. % Weak links	17.1	11.4	−0.22	−0.12	0.14	

Table 3. Correlations matrix for high-performing groups.

	Mean	SD	1	2	3	4
1. Group heterogeneity	5.75	0.493				
2. Tolerance of differences	4.51	0.545	−0.33			
3. Task analyzability	4.51	0.577	0.27	−0.18		
4. % Weak links	13.3	6.7	0.29	−0.27	0.23	

Table 4. Predicting the proportion of the technical advice network that consists of weak external links for *low-performing groups*.

Variable	% Weak links	Mean	SD
1. Group heterogeneity	−144.86	5.69	0.326
2. Tolerance of differences	−240.46**	4.28	0.452
3. Task analyzability	67.75	4.14	0.726
1. Group heterogeneity * 3. Task analyzability	−11.14		
1. Group heterogeneity * 2. Tolerance of differences	41.18**		
Constant	851.55		
R^2	0.32		
Adjusted R^2	0.14		
$N = 24$			

Notes: $*p < 0.1$ and $**p < 0.05$.

Table 5. Predicting the proportion of the technical advice network that consists of weak external links for *high-performing groups*.

Variable	% Weak links	Mean	SD
1. Group heterogeneity	115.13**	5.75	0.493
2. Tolerance of differences	89.49**	4.51	0.454
3. Task analyzability	52.34**	4.51	0.577
1. Group heterogeneity * 3. Task analyzability	−8.87**		
1. Group heterogeneity * 2. Tolerance of Differences	−16.30**		
Constant	−635.94		
R^2	0.38**		
Adjusted R^2	0.27**		
$N = 25$			

Notes: $*p < 0.1$ and $**p < 0.05$.

that consists of weak boundary-spanning links to the level of analyzability of the task. The only interaction term that is statistically significant for low-performing groups is for "heterogeneity * tolerance of differences." For high-performing groups, Table 5 indicates that all the hypothesized interaction terms are statistically significant.

Also run, but not shown, was a test of whether the increment in the proportion of variance accounted for by the interaction terms, when compared with just the main effects, was statistically significant. For high-performing groups, the proportion of variance accounted for by the interaction terms was significant at the 0.05 level.

The use of multiple regression to analyze *interaction effects* is straightforward and is covered in many statistical texts and scholarly articles [Cohen and Cohen (1983, pp. 320–325); Marsden (1981); Stone (1988)]. Cohen and Cohen [1983, pp. 322–324] describe how interaction effects can be interpreted in conjunction with the main effects of the interacting variables by rearranging the regression equation so that the conditionality of the effects of a focal variable on other variables is clear.

Selected lines can then be plotted to illustrate the conditional formulation that is implicit in the interaction terms: the regression of Y (i.e. the proportion of weak boundary-spanning technical advice links) on task analyzability depends upon the value of a moderating variable such as group heterogeneity. Cohen and Cohen [1983, p. 323] recommend three representative members: one for a "low" value of a moderating variable (one standard deviation below the mean), one for an "average" value of a moderating variable (at the mean), and one for a "high" value of a moderating variable (one standard deviation above the mean).

Figure 4 shows the influence of group heterogeneity and tolerance of differences upon the proportion of the technical advice network that consists of weak boundary-spanning technical advice links. In Fig. 4, tolerance of differences has been held constant at its low value (−1 S.D.) and the values of task analyzability vary from high to low. In Fig. 5, tolerance of differences has been held constant at its high value (+1 S.D.) and the values of task analyzability vary from high (i.e. task-related problems are easy to analyze) to low (i.e. task-related problems are difficult to analyze).

The hypotheses previously presented can be tested by observing the direction of the changes that occur in the proportion of weak boundary-spanning technical advice links (i.e. whether increasing or decreasing) for a specified value of a moderating variable as the value of the focal variable (i.e. task analyzability) is increased from high to low, and all other variables in the regression equation are held constant at their average values. For example, the interaction term between group heterogeneity and task analyzability can be tested by determining if the slope of the line for each value (i.e. low, medium, and high) of a moderator variable (i.e. group heterogeneity) is positive or negative as hypothesized.

6. Results

6.1. *The role of boundary-spanning technical advice links*

The results for high-performing groups are shown graphically in Figs. 3–5. Figure 3 shows the influence of task analyzability and group heterogeneity on the proportion of the technical advice network that consists of weak boundary-spanning links. In

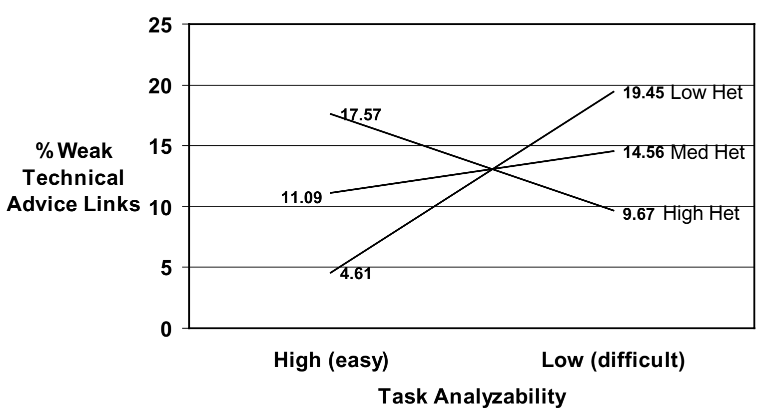

Fig. 3. Interaction of analyzability and heterogeneity, with tolerance of differences held constant at its mean value.

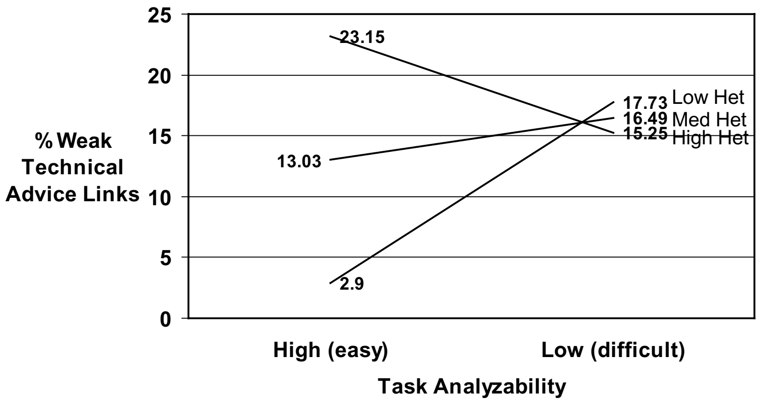

Fig. 4. Interaction of analyzability and heterogeneity, with tolerance of differences held constant at one standard deviation below its mean.

this figure, tolerance of differences has been held constant at its *average* value and the values of task analyzability vary over their observed range: from high (task-related problems are easy to analyze) to low (task-related problems are difficult to analyze), a numerical range from around 3 to 6.

As shown in Fig. 3, the slopes are in the hypothesized direction, thereby lending support to hypotheses 1A and 1B. An examination of Fig. 3 indicates that when group heterogeneity is perceived to be low, a decrease in task analyzability from high (i.e. task-related problems are easy to analyze) to low (i.e. task-related problems are difficult to analyze) results in an *increase* in the proportion of the group's technical advice network that consists of weak boundary-spanning links. When heterogeneity is perceived to be high, a decrease in task analyzability from high (i.e. task-related problems are easy to analyze) to low (i.e. task-related problems are difficult to analyze) results in a *decrease* in the proportion of the project group's technical advice network that consists of weak boundary-spanning links. Thus, hypotheses 1A and 1B are supported.

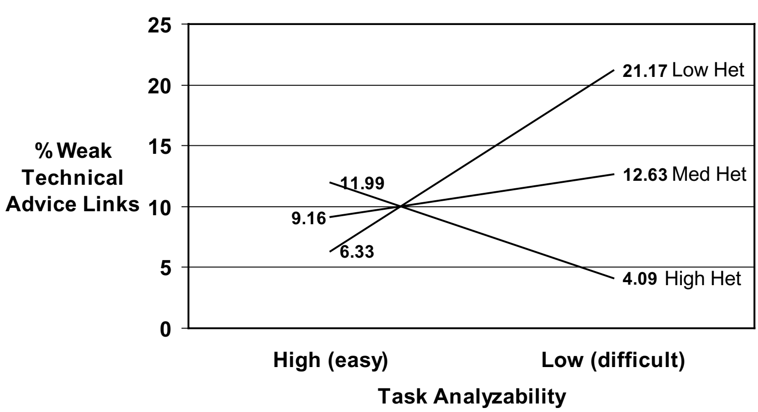

Fig. 5. Interaction of analyzability and heterogeneity, with tolerance of differences held constant at one standard deviation above its mean.

Figure 4 shows the influence of tolerance of differences and group heterogeneity upon the proportion of a group's technical advice network that consists of weak boundary-spanning links. In Fig. 4, tolerance of differences has been held constant at its low value (-1 S.D.) and the values of task analyzability vary from high (i.e. task-related problems are easy to analyze) to low (i.e. task-related problems are difficult to analyze). Figure 5 is similar to Fig. 4, except that now tolerance of differences has been held constant at its high value ($+1$ S.D.).

Comparing Fig. 4 (tolerance of differences is low) with Fig. 5 (tolerance of differences is high), one can see that when heterogeneity is high an increase in tolerance of differences *decreases* the proportion of the group's technical advice network that consists of weak boundary-spanning links. When heterogeneity is low, an increase in tolerance of differences *increases* the proportion of the group's technical advice network that consists of weak boundary-spanning links. Note that the slopes of the lines in these two charts do not change due to an increase in tolerance of differences, since no interaction term was hypothesized between "tolerance of differences" and "task analyzability." Thus, there is support for hypotheses 2A and 2B.

Figures 3 through 5 indicate that when task-related problems are *difficult* to analyze, the proportion of the group's technical advice network that consists of weak boundary-spanning technical advice links is at its maximum when a group has a high tolerance for difference and group heterogeneity is perceived to be low, and the proportion of weak boundary-spanning links is at a minimum when a group has a high tolerance for differences and group heterogeneity is perceived to be high. These results seem to indicate that when tolerance of differences is high, high-performing groups use their internal group heterogeneity as the diverse information base needed to deal with task-related problems that are difficult to analyze, but rely on weak boundary-spanning technical advice links to supply the diverse information needed if the group is homogeneous and therefore probably lacking the internal diversity needed to deal with task-related problems that are difficult to analyze.

Figures 3 through 5 also indicate that when task analyzability is high (i.e. task-related problems are *easy* to analyze), the proportion of the group's technical advice network that consists of weak boundary-spanning links is at its maximum when a group has a low tolerance for differences and group heterogeneity is high and is at a minimum when a group has a low tolerance for differences and group heterogeneity is low. These results seem to indicate that when tolerance of differences is low, high-performing project groups may rely on weak boundary-spanning technical advice links in order to avoid the potential for conflict that is created when heterogeneity is high.

Note also that a change in tolerance of differences from low to high has a relatively large impact on the proportion of the group's technical advice network that consists of weak boundary-spanning technical advice links when group heterogeneity is perceived to be high. When heterogeneity is perceived to be low, a change in tolerance of differences from low to high has a fairly modest influence on the portion of weak boundary-spanning technical advice links. Thus, it seems that for these high-performing groups, a climate that is tolerant of differences is key when a group wants to tap the technical advice that should be available within a heterogeneous group in order to deal with difficult-to-analyze task-related problems, but when such tolerance of differences is lacking the group may compensate by relying more on weak boundary-spanning links for any technical advice needed.

For low-performing groups, the interaction term between heterogeneity and task analyzability is not statistically significant. Although the interaction term between heterogeneity and tolerance of differences is statistically significant, the slopes are not in the hypothesized direction. Indeed, an examination of this interaction term reveals a pattern almost the opposite of that hypothesized for high-performing groups. The results indicate that high-performing project groups are able to find the diverse information needed to successfully deal with the difficult-to-analyze task-related problems facing the group whereas low-performing groups are not able to do so. Thus, there is support for hypotheses 1C and 2C.

7. Discussion and Conclusions

7.1. *Discussion of results*

The results of this study indicate that high-performing groups adjust their proportion of weak intra-organizational boundary-spanning technical advice links to import the diverse information needed to deal with difficult-to-analyze task-related problems. For the high-performing project groups in this study, the proportion of weak boundary-spanning technical advice links depends not only on the extent to which task-related problems are analyzable but also on the heterogeneity and tolerance of the group. This is not the case for low-performing groups.

The results of this study may contribute to understanding some of the conflicting evidence concerning the impact of group heterogeneity [Ancona and Caldwell (1992a); Bantel and Jackson (1989); Guzzo and Dickson (1996); Jackson (1992); Zenger and Lawrence (1989)]. Project groups may adjust their proportion of boundary-spanning technical advice links in order to change the diversity of the information base. Thus, not all project groups would need the same proportion of weak boundary-spanning technical advice links. Even if several project groups faced tasks with the same degree of task analyzability, each of these groups may have a different proportion of weak boundary-spanning technical advice links, depending on the perceived heterogeneity of the group. That is, one project group may have a greater proportion of weak boundary-spanning links than another group not because tasks are perceived to be difficult to analyze, but because the heterogeneity of the group is perceived to be lower.

Another possible reason for some of the conflicting evidence concerning the impact of group heterogeneity is that group climate may influence how heterogeneity is perceived by group members. The climate of the group is important since it is easier to approach heterogeneous co-workers for technical advice when tolerance of differences is high. As shown in Figs. 3 and 5, tolerance of differences has a fairly large influence on the proportion of weak boundary-spanning technical advice links when heterogeneity is high. When heterogeneity is low, a change in tolerance of differences from low to high has only a moderate influence on the proportion of weak boundary-spanning technical advice links.

Drach-Zahavy and Somech [2010, p. 157] propose that "...a team member who perceives himself or herself as dissimilar from the other team members will find it even easier to create interpersonal contacts with members of other teams." Zenger and Lawrence [1989], Choi [2002], and Keller [2001] suggest that heterogeneous groups may have a higher rate of communication with individuals outside the group. As shown in Fig. 4, this may only be the case when tasks are easy to analyze and the climate of the group does not encourage interaction with heterogeneous co-workers within the group.

The only interaction term that is statistically significant for low-performing groups is "heterogeneity * tolerance of differences." It should be noted that the sign of this interaction term is the *opposite* of that for high-performing groups. One possible explanation is that, for these low-performing groups, a high tolerance of differences may allow group members to avoid interactions with heterogeneous co-workers and to keep to themselves. Thus, it could be that low-performing groups may seek to avoid new and diverse information instead of embracing it as an aid to the problem-solving process.

The results of this study may have implications related to the design and use of heterogeneous, self-contained project groups in organizations as a way to encourage the development of innovative and creative solutions to task-related problems. If, indeed, autonomous, self-contained groups such as "skunk works" are successful at developing innovative solutions to difficult-to-analyze task-related problems, it may be because these groups are both highly heterogeneous and highly tolerant of others with differing viewpoints. Such groups would minimize their need for weak boundary-spanning technical advice links (see Appendix A, note 4).

The focus of this study has been on boundary-spanning activities between teams and other members within the same organization, thereby excluding boundary-spanning activities to individuals external to the organization. The open innovation paradigm [Chesbrough (2003)] predicts that the best innovation performance occurs among firms that balance internal and external sources of information even though the emphasis seems to be more on external ties. The findings of our research suggest that one cannot exclude internal sources of innovation ideas if they come from other groups within the organization, especially if departure from incremental innovation is of interest [Cohen *et al.* (1996)]. One of the emerging central questions guiding open innovation research streams is where to draw the boundary between the organization's focal problems, issues, challenges, and innovative process in pursuit of new ideas. The vast majority of the research questions since 1993 and Chesbrough's book establish the boundary as the external environment and context of the firm. Although recent research has been concentrated on studies of open innovation and consequently, there are few recent studies of boundary spanning within an organization. In the current research, we bring the focus back to boundary spanning within the large incumbent firm. We examine and establish statistically significant tendencies showing how important internal boundary-spanning is for incumbent, engineering-intensive firms.

7.2. *Limitations and future research*

This study has concentrated upon the ability of weak boundary-spanning technical advice links to provide project groups with the diverse information needed to deal with difficult-to-analyze task-related problems. Additional research is needed to replicate this study with additional groups in other types of organizations before it can be determined if the results described here are generalizable. In particular, the project groups investigated in this study were tasked with making improvements to existing but highly complex systems, not with developing completely new products or systems. Thus, these results may not apply to new product development groups. If additional research does support the current study, it may explain why some empirical studies have found that heterogeneous groups are positively associated with group performance, whereas other studies have found that heterogeneous groups are negatively associated with team performance [Williams and O'Reilly (1998)].

Burt [1992, 2004] has argued that it is not so much the strength of a tie that determines its ability to transfer non-redundant information, but whether a structural hole exists between an individual's contacts. A structural hole exists if an individual's weak ties are not connected to each other or connected to that individual's strong-tie contacts. Unfortunately, the ego-centric-based network data collected for this study do not allow one to determine whether it is structural holes, as Burt argues, or tie strength, as Granovetter [1973] proposes, that will provide access to new and non-redundant information.

Our basic argument is that high-performing groups have the ability to find and process the information needed to successfully deal with the task-related problems whereas low-performing groups are not able to do so. A limitation of this study is the ambiguity concerning the direction of causality. Is the direction of causality from low performance due to a project group avoiding seeking information or is it that not seeking diverse information causes the low performance?

A number of future research questions emerge from these results. First, there is an implied connection between the individual and group level of aggregation in this research stream that warrants exploring. That is, there is a continuing interest in boundary-spanning roles rekindled by the open innovation movement, and findings show relationships between internal extra-group activities. For example, Ettlie and Elsenbach [2007] found that the traditional role of the R&D group supervisor as the prime source of the gatekeeper role no longer applies in both U.S. and German idea sourcing for new products and services. This role is now distributed throughout the technical ranks in firms and, especially in SMEs (small- and medium-sized firms), boundary-spanning often is done by senior technical managers. Miller *et al.* [2007] found that ideas originating outside R&D groups are more likely to lead to radical departures from current technical trajectories. This effect needs to be incorporated in this line of research. The nature of the task itself, in a qualitative way, might also be taken into account regardless of its analyzability. For example, is it a fast-track project? Are partners involved?

Third, does crossing different boundaries (e.g. project group, division, department, organization) result in different types of information and thus, different approaches to problem solving? Individuals in different work groups and in different organizations are likely to have different types of diverse information. Which boundary should they cross to find the diverse information needed?

Fourth, one extension of this study would be to ask the simple research question of how it is that individuals are nominated to fill the roles that compensate for a lack of heterogeneity in a group. Do these individuals have broader, external experience from job rotation? Have they recently arrived in this firm or group assignment? Which boundary (internal or external to the organization) is more important? The open innovation literature has largely overlooked the issue of which boundary to cross for effective information sharing.

Fifth, and finally, this study has focussed on weak technical advice links going outside the project group but inside the organization. Future studies should examine the role of strong and duplex links and their influence on project team performance. For example, what are the advantages of a duplex link that is used for technical advice but the link also has a social component?

Our focus was on technical advice links outside the project group but inside the organization studied. Future studies should include data on technical advice links going outside the focal organization. Our major contribution in reporting these significant internal intra-organizational relationships is to help re-establish boundary-spanning as a central construct in the literature on engineering project group performance. Specifically, internal, structural characteristics of a firm appear to have a strong influence on how project groups deal with difficult-to-analyze task-related problems. The way internal and external boundaries are established and studied in organizational research is important to organizing future research in this field. In spite of the limitations of the current study, whereby we did not collect information on links external to the firm, such as sourcing technical advice from suppliers, the strength of intra-organizational sources of technical advice is an important finding.

References

Allen, T. J. (1977). *Managing the Flow of Technology: Technology Transfer and the Dissemination of Technological Information within the R&D Organization*. MIT Press, Cambridge, MA.

Ancona, D. G. and Caldwell, D. F. (1992a). Bridging the boundary: External process and performance in organizational teams. *Administrative Science Quarterly*, **37**: 634–665.

Ancona, D. G. and Caldwell, D. F. (1992b). Demography and design: Predictors of new product team performance. *Organization Science*, **3**: 321–341.

Ashkanasy, N. M., Wilderom, C. P. M. and Peterson, M. F. (Eds.) (2000). *Handbook of Organizational Culture and Climate*. Sage Publications, Thousand Oaks, CA.

Baker, N. R., Winkofsky, E. P., Langmeyer, L. and Sweeney, D. J. (1980). Idea generation: A procrustean bed of variables, hypotheses, and implications. *Management of Research and Innovation*, eds. B. V. Dean and J. L. Goldhar. North Holland, NY, pp. 33–51.

Bantel, K. A. and Jackson, S. E. (1989). Top management and innovations in banking: Does the composition of the top team make a difference? *Strategic Management Journal*, **10**: 107–124.

Berchicci, L. (2013). Towards an open R&D system: Internal R&D investment, external knowledge acquisition and innovative performance. *Research Policy*, **42**: 117–127.

Bowers, C. A., Pharmer, J. A. and Salas, E. (2000). When member homogeneity is needed in work teams: A meta-analysis. *Small Group Research*, **31**: 305–327.

Brass, D. J., Butterfield, K. D. and Skaggs, B. C. (1998). Relationships and unethical behavior: A social network perspective. *Academy of Management Review*, **23**: 14–31.

Brewer, M. B. (1979). In-group bias in the minimal intergroup situation: A cognitive-motivational analysis. *Psychological Bulletin*, **86**: 307–324.

Burt, R. S. (1983). Distinguishing relational contents. *Applied Network Analysis*, eds. R. S. Burt and M. J. Minor. Sage Publications, Beverly Hills, CA.

Burt, R. S. (1992). *Structural Holes: The Social Structure of Competition*. Harvard University Press, Cambridge, MA.

Burt, R. S. (2004). Structural holes and good ideas. *American Journal of Sociology*, **110**: 349–399.

Cammann, C., Fichman, M., Jenkins, G. D. and Klesh, J. K. (1983). Assessing the attitudes and perceptions of organizational members. *Assessing Organizational Change*, eds. S. E. Seashore, E. E. Lawler, P. H. Mirvis and Cammann, C., John Wiley & Sons, NY, pp. 71–137.

Chatman, J. A. and Flynn, F. J. (2001). The influence of demographic heterogeneity on the emergence and consequences of cooperative norms in work teams. *Academy of Management Journal*, **44**: 956–974.

Chatman, J. A., Polzer, J. T., Barsade, S. G. and Neale, M. A. (1998). Being different yet feeling similar: The influence of demographic composition on work processes and outcomes. *Administrative Science Quarterly*, **43**: 749–780.

Chesbrough, H. (2003). *Open Innovation*. Harvard Business School Press, Boston.

Chi, L., Liu, Q. and Yu, X. (2013). Effects of top management team heterogeneous background and behavioral attributes on the performance of new ventures. *Systems Research and Behavioral Science*, **30**: 354–368.

Choi, J. N. (2002). External activities and team effectiveness: Review and theoretical development. *Small Group Research*, **33**: 181–208.

Cohen, J. and Cohen, P. (1983). *Applied Multiple Regression/Correlation Analysis for the Behavioral Sciences*, 2nd edn. Lawrence Erlbaum Associates, Hillsdale, New Jersey.

Cohen, M. A., Eliashbert, J. and Ho, T. H. (1996). New product development: The performance and time to market trade-off. *Management Science*, **42**, 2: 173–186.

Cronin, M. A. and Weingart, L. R. (2007). Representational gaps, information processing, and conflict in functionally diverse teams. *Academy of Management Review*, **32**: 761–773.

Cummings, J. N. (2004). Work groups, structural diversity, and knowledge sharing in a global organization. *Management Science*, **50**: 352–364.

Daft, R. L. and Lengel, R. H. (1984). Information richness: A new approach to managerial behavior and organizational design. *Research in Organizational Behavior*, eds. B. Staw and L. L. Cummings. Jai Press, Greenwich, CT, pp. 191–233.

Daft, R. L. and Lengel, R. H. (1986). Organizational information requirements, media richness and structural design. *Management Science*, **32**: 554–571.

Daft, R. L. and Macintosh, N. B. (1981). A tentative exploration into the amount and equivocality of information processing in organizational work units. *Administrative Science Quarterly*, **26**: 207–224.

Dahlander, L. and Gann, D. M. (2010). How open is innovation. *Research Policy*, **39**: 699–709.

Dahlander, L. and McFarland, D. A. (2013). Ties that last: The formation and persistence of research collaborations over time. *Administrative Science Quarterly*, **58**: 69–110.

Dokko, G., Kane, A. and Tortoriello, M. (2014). One of us or one of my friends: How social identity and tie strength shape the and creative generativity of boundary-spanning ties. *Organization Studies*, **35**: 703–726.

Drach-Zahavy, A. and Somech, A. (2010). From an intrateam to an interteam perspective of effectiveness: The role of interdependence and boundary activities. *Small Group Research*, **41**: 143–174.

Ebersberger, B., Bloch, C., Herstad, S. and Van de Velde, E. (2012). Open innovation practices and their effect on innovation performance. *International Journal of Innovation and Technology Management*, **9**: 23pp.

Ely, R. J. and Thomas, D. A. (2001). Cultural diversity at work: The effects of diversity perspectives on work group processes and outcomes. *Administrative Science Quarterly*, **46**: 229–273.

Ettlie, J. and Elsenbach, J. (2007). The changing role of R&D gatekeepers, *Research-Technology Management*, **50**, 5: 59–66.

Ettlie, J. and Rosenthal, S. (2012). Service innovation in manufacturing. *Journal of Services Management*, **23**, 3: 440–454.

Falcione, R. L., Sussman, L. and Herden, R. P. (1987). Communication climate in organizations. *Handbook of Organizational Communication*, eds. F. R. Jablin, L. L. Putnam, K. Roberts and L. Porter. Sage Publications, Newbury Park, CA, pp. 195–227.

Faraj, S. and Yan, A. (2009). Boundary work in knowledge teams. *Journal of Applied Psychology*, **94**: 604–617.

Fischer, W. A. (1980). Scientific and technical information and the performance of R&D groups. *Management of Research and Innovation*, eds. B. V. Dean and J. L. Goldhar. North Holland, NY, pp. 67–89.

Fleming, L., Mingo, S. and Chen, D. (2007). Collaborative brokerage, generative creativity, and creative success. *Administrative Science Quarterly*, **52**: 443–475.

Fletcher, D. (1998). Swimming around in their own ponds: The weakness of strong ties in developing innovative practices. *International Journal of Innovation Management*, **2**: 137–159.

Frenzen, J. and Nakamoto, K. (1993). Structure, cooperation, and the flow of market information. *Journal of Consumer Research*, **20**: 360–375.

Gales, L. and Mansour-Cole, D. (1995). User involvement in innovation projects: Toward an information processing model. *Journal of Engineering and Technology Management*, **12**: 77–109.

Gales, L., Porter, P. and Mansour-Cole, D. (1992). Innovation project technology, information processing and performance: A test of the Daft and Lengel conceptualization. *Journal of Engineering and Technology Management*, **9**, 3–4: 303–338.

Granovetter, M. S. (1973). The strength of weak ties. *American Journal of Sociology*, **78**: 1360–1380.

Granovetter, M. S. (1982). The strength of weak ties: A network theory revisited. *Social Structure and Network Analysis*, eds. P. V. Marsden and N. Lin. Sage Publications, Beverly Hill, CA, pp. 105–130.

Guzzo, R. A. and Dickson, M. W. (1996). Teams in organizations: Recent research on performance and effectiveness. *Annual Review of Psychology*. Annual Reviews, Palo Alto, CA.

Hansen, M. T. (1999). The search-transfer problem: The role of weak ties in sharing knowledge across organizational subunits. *Administrative Science Quarterly*, **44**: 82–111.

Harrison, D. A. and Klein, K. J. (2007). What's the difference? Diversity constructs as separation, variety, or disparity in organizations. *Academy of Management Review*, **32**: 1199–1228.

Henderson, R. M. and Clark, K. B. (1990). Architectural innovation: The reconfiguration of existing product technologies and the failure of established firms. *Administrative Science Quarterly*, **35**: 9–30.

Hobman, E. V., Bordia, P. and Gallois, C. (2004). Perceived dissimilarity and work group involvement. *Group and Organization Management*, **29**: 560–587.

Hoffman, L. R. (1959). Homogeneity of member personality and its effects on group problem solving. *Journal of Abnormal and Social Psychology*, **58**: 28–32.

Hoffman, L. R., Harburg, E. and Maier, N. R. F. (1962). Differences and disagreement as factors in creative group problem solving. *Journal of Abnormal and Social Psychology*, **64**: 206–214.

Hoffman, L. R. and Maier, N. R. F. (1961). Quality and acceptance of problem solutions by members of homogeneous and heterogeneous groups. *Journal of Abnormal and Social Psychology*, **62**: 401–407.

Homan, A. C., van Knippenbeg, D., Van Kleef, G. A. and De Dreu, C. K. W. (2007). Bridging faultlines by valuing diversity: Diversity beliefs, information elaboration, and performance in diverse work groups. *Journal of Applied Psychology*, **92**: 1189–1199.

Horwitz, S. K. and Horwitz, I. B. (2007). The effects of team diversity on team outcomes: A meta-analytic review of team demography. *Journal of Management*, **33**: 987–1015.

Ibarra, H. (1995). Race, opportunity, and diversity of social circles in managerial networks. *Academy of Management Journal*, **38**: 673–703.

Ito, J. K. and Peterson, R. B. (1986). Effects of task difficulty and interunit interdependence on information processing systems. *Academy of Management Journal*, **39**: 139–149.

Jackson, S. E. (1992). Team composition in organizations. *Group Process and Productivity*, eds. S. Worchel, W. Wood and J. Simpson. Sage Publishers, Beverly Hills, CA, pp. 138–176.

Jackson, S. E. and Joshi, A. (2011). Work team diversity. *APA Handbook of Industrial and Organizational Psychology*, ed. S. Zedeck. Vol. 1. APA, Washington, D.C., pp. 651–686.

Jackson, S. E., May, K. E. and Whitney, K. (1995). Understanding the dynamics of diversity in decision-making teams. *Team Effectiveness and Decision Making in Organizations*, eds. R. A. Guzzo and E. Salas. Jossey-Bass, San Francisco, pp. 383–396.

Jehn, K. A. (1995). A multimethod examination of the benefits and detriments of intragroup conflict. *Administrative Science Quarterly*, **40**: 256–282.

Jehn, K. A., Northcraft, G. B. and Neale, M. A. (1999). Why differences make a difference: A field study of diversity, conflict and performance in workgroups. *Administrative Science Quarterly*, **44**: 741–763.

Johnston, R. and Gibbons, M. (1975). Characteristics of information usage in technological innovation. *IEEE Transactions on Engineering Management*, **EM-22**: 27–34.

Joshi, A. (2006). The influence of organizational demography on the external networking behavior of teams. *Academy of Management Review*, **31**: 583–595.

Joshi, A., Pandey, N. and Han, G. (2009). Bracketing team boundary-spanning: An examination of task-based, team-level, and contextual antecedents. *Journal of Organizational Behavior*, **30**: 731–759.

Kasperson, C. J. (1978). An analysis of the relationship between information sources and creativity in scientists and engineers. *Human Communication Research*, **4**: 113–119.

Kenis, P. and Knoke, D. (2002). How organizational field networks shape interorganizational tie-formation rates. *Academy of Management Review*, **27**: 275–293.

Kenny, D. A. and La Voie, L. (1985). Separating individual and group effects. *Journal of Personality and Social Psychology*, **48**: 339–348.

Keller, R. T. (2001). Cross-functional project groups in research and new product development: Diversity, communications, job stress, and outcomes. *Academy of Management Journal*, **44**: 547–555.

King, E. B., Hebel, M. R. and Beal, D. J. (2009). Conflict and cooperation in diverse workgroups. *Journal of Social Issues*, **65**: 261–285.

Kohler, C., Sofka, W. and Grimpe, C. (2012). Selective search, sectoral patterns and the impact on product innovation performance. *Research Policy*, **41**: 1344–1356.

Krackhardt, D. (1992). The strength of strong ties: The importance of philos in organizations. *Networks and Organization*, eds. N. Nohria and R. G. Eccles. Harvard Business School Press, Boston, MA, pp. 216–239.

Lawrence, B. S. (1997). The back box of organizational demography. *Organization Science*, **8**: 1–22.

Lengel, R. H. and Daft, R. L. (1988). The selection of communication media as an executive skill. *Academy of Management Executive*, **2**, 3: 225–232.

Levin, D. Z., Walter, J., Appleyard, M. M. and Cross, R. (2016). Relational enhancement: How the relational dimension of social capital unlocks the value of network-bridging ties. *Group and Organization Management*, **41**, 4: 415–457.

Mannix, E. and Neale, M. A. (2005). What differences make a difference? The promise and reality of diverse teams in organizations. *Psychological Science in the Public Interest*, **6**: 31–55.

Marrone, J. A., Tesluk, P. E. and Carson, J. B. (2007). A multi-level investigation of antecedents and consequences of team member boundary-spanning behavior. *Academy of Management Journal*, **50**: 1423–1439.

Marsden, P. V. (1981). Conditional effects in regression analysis. *Linear Models in Social Research*, ed. P. V. Marsden. Sage Publications, Beverly Hills, CA, pp. 97–116.

Marsden, P. V. and Campbell, K. E. (1984). Measuring tie strength. *Social Forces*, **63**: 482–501.

Marsden, P. V. and Campbell, K. E. (2012). Reflections on conceptualizing and measuring tie strength. *Social Forces*, **91**: 17–23.

McEvily, B. and Zaheer, A. (1999). Bridging ties: A source of firm heterogeneity in competitive capabilities. *Strategic Management Journal*, **20**: 1133–1156.

Menzel, H. (1962). Planned and unplanned scientific communication. *The Sociology of Science*, eds. B. Barber and W. Hirsch. Greenwood Press, Westport, CT, pp. 417–441.

Michelfelder, I. and Kratzer, J. (2013). Why and how combining strong and weak ties within a single interorganizational R&D collaboration outperforms other collaboration structures. *Journal of Product Innovation Management*, **30**: 1159–1177.

Michelino, F., Lamberti, E., Cammarano, A. and Caputo, M. (2015). Measuring open innovation in the bio-pharmaceutical industry. *Creativity and Innovative Management*, **24**: 4–28.

Miles, R. H. (1980). *Macro Organizational Behavior*. Goodyear Publishing Company, Santa Monica, CA.

Milliken, F. J. and Martins, L. L. (1996). Searching for common threads: Understanding the multiple effects of diversity in organizational groups. *The Academy of Management Review*, **21**: 402–433.

Minshall, T., Kouris, S., Mortara, L., Schmithausen, P. and Weiss, D. (2014). Developing infrastructure to support open innovation: Case studies for the east of England. *International Journal of Innovation and Technology Management*, **11**: 21pp.

Mitchell, J. C. (1987). The components of strong ties among homeless women. *Social Networks*, **9**: 37–47.

Moran, P. (2005). Structural vs. relational embeddedness: Social capital and managerial performance. *Strategic Management Journal*, **26**: 1129–1151.

Mors, M. L. (2010). Innovation in a global consulting firm: When the problem is too much diversity. *Strategic Management Journal*, **31**: 841–872.

Nishii, L. A. (2013). The benefits of climate for inclusion for gender-diverse groups. *Academy of Management Journal*, **56**: 1754–1774.

Oh, H., Labianca, G. and Chung, M. (2006). A multilevel model of group social capital. *Academy of Management Review*, **31**: 569–582.

Olivera, P. and von Hippel, E. (2011). Users as service innovators: The case of banking services. *Research Policy*, **40**: 806–818.

Pelled, L. H., Eisenhardt, K. M. and Xin, K. R. (1999). Exploring the black box: An analysis of work group diversity, conflict, and performance. *Administrative Science Quarterly*, **44**: 1–28.

Pelz, D. C. and Andrews, F. (1976). *Scientists in Organizations* (Revised Ed.). Institute for Social Research, University of Michigan, Ann Arbor, MI.

Perrow, C. (1967). A framework for the comparative analysis of organizations. *American Sociological Review*, **32**: 194–208.

Perrow, C. (1970). *Organizational Analysis: A Sociological View*. Wadsworth Publishing Company, Belmont, CA.
Perry-Smith, J. E. (2006). Social yet creative: The role of social relationships in facilitating individual creativity. *Academy of Management Journal*, **49**: 85–101.
Perry-Smith, J. E. and Shalley, C. E. (2003). The social side of creativity: A static and dynamic social network prospective. *Academy of Management Review*, **49**: 85–101.
Ray, E. B. and Miller, K. I. (1991). The influence of communication structure and social support on job stress and burnout. *Management Communication*, **4**: 506–527.
Reagans, R. and McEvily, B. (2003). Network structure and knowledge transfer: The effects of cohesion and range. *Administrative Science Quarterly*, **48**: 240–267.
Reagans, R. and Zuckerman, E. W. (2001). Networks, diversity, and productivity: The social capital of corporate R&D teams. *Organization Science*, **12**: 502–517.
Reagans, R., Zuckerman, E. W. and McEvily, B. (2004). How to make the team: Social networks vs. demography as criteria for designing effective teams. *Administrative Science Quarterly*, **49**: 101–133.
Rost, K. (2011). The strength of strong ties in the creation of innovation. *Research Policy*, **40**: 588–604.
Schneider, B. (1990). *Organizational Climate and Culture*. Jossey-Bass, San Francisco, CA.
Schotter, A. and Beamish, P. W. (2011). Performance effects of MNC headquarter-subsidiary conflict and the role of boundary spanners: The case of headquarter initiative rejection. *Journal of International Management*, **17**: 243–259.
Siegel, S. M. and Kaemmerer, W. F. (1978). Measuring the perceived support for innovation. *Journal of Applied Psychology*, **63**: 553–562.
Slepian, J. L. (2013). Cross-functional teams and organizational learning: A model and cases from telecommunications operating companies. *International Journal of Innovation and Technology Management*, **10**: 1350005-1–135005-29.
Smith, C. G. (1970). Consultation and decision processes in a research and development conlaboratory. *Administrative Science Quarterly*, **15**: 203–215.
Smith, C. G. (1971). Scientific performance and the composition of research teams. *Administrative Science Quarterly*, **16**: 486–495.
Smith, K. G., Collins, C. J. and Clark, K. D. (2005). Existing knowledge, knowledge creation capability, and the rate of new product introduction in high-technology firms. *Academy of Management Journal*, **48**: 346–357.
Sofka, W. and Grimpe, C. (2010). Specialized search and innovation performance — Evidence across Europe. *R&D Management*, **40**: 310–323.
Somech, A. and Drach-Zahavy, A. (2013). Translating team creativity to innovation implementation: The role of team composition and climate for innovation. *Journal of Management*, **39**: 684–708.
Somech, A. and Khalaili, A. (2014). Team boundary activity: Its mediating role in the relationship between structural conditions and team innovation. *Group and Organizational Management*, **39**: 274–299.
Sosa, M. E. (2011). Where do creative interactions come from? The role of tie content and social networks. *Organization Science*, **22**: 1–21.
Stone, E. F. (1988). Moderator variables in research: A review and analysis of conceptual and methodological issues. *Research in Personnel and Human Resources Management*, eds. G. R. Ferris and K. M. Rowland, Vol. 6. JAI Press, Greenwich, CT.
Tichy, N. M., Tushman, M. L. and Fombrun, C. (1979). Social network analysis for organizations. *Academy of Management Review*, **4**: 507–519.
Tiwana, A. (2008). Do bridging ties complement strong ties? An empirical examination of alliance ambidexterity. *Strategic Management Journal*, **29**: 251–272.
Triandis, H. E., Hall, E. R. and Ewen, R. B. (1965). Member heterogeneity and dyadic creativity. *Human Relations*, **18**: 33–55.

Tushman, M. L. (1977). Special boundary roles in the innovation process. *Administrative Science Quarterly*, **22**: 587–605.

Tushman, M. L. (1979). Work characteristics and subunit communication structure: A contingency analysis. *Administrative Science Quarterly*, **24**: 82–89.

Tushman, M. L. and Nadler, D. A. (1978). Information processing as an integrating concept in organizational design. *Academy of Management Review*, **3**: 613–624.

Uzzi, B. (1996). The sources and consequences of embeddedness for the economic performance or organizations: The network effect. *American Sociological Review*, **61**: 674–698.

Van de Ven, A. and Ferry, H. D. (1980). *Measuring and Assessing Organizations.* John Wiley & Sons, NY.

van Knippenberg, D. and Schippers, M. C. (2007). Work group diversity. *Annual Review of Psychology*, eds. M. I. Posner and M. K. Rothbard, Vol. 58, Annual Reviews, Palo Alto, CA, pp. 515–541.

van Knippenberg, D., DeDreu, C. K. W. and Homan, A. C. (2004). Work group diversity and group performance: An integrative model and research agenda. *Journal of Applied Psychology*, **89**: 1008–1022.

van Knippenberg, D., Haslam, S. A. and Platow, M. J. (2007). Unity through diversity: Value-in-diversity beliefs, work group diversity, and group identification. *Group Dynamics: Theory, Research, and Practice*, **11**: 207–222.

Verbrugge, L. M. (1979). Multiplexity in adult friendships. *Social Forces*, **57**: 1286–1309.

Wan, H. and Quan, X. I. (2014). Toward a framework of the process of open innovation — Case of Acclarent in the medical Device Industry. *International Journal of Innovation and Technology Management*, **11**: 15pp.

Wang, C., Sung, H., Chen, D. and Huang, M. (2017). Strong ties and weak ties of the knowledge spillover network in the semiconductor industry. *Technological Forecasting and Social Change*, **118**: 114–127.

Wanous, J. P. and Youtz, M. A. (1986). Solution diversity and the quality of group decisions. *Academy of Management Journal*, **29**: 149–159.

Wasserman, S. and Faust, K. (1994). *Social Network Analysis.* Cambridge University Press, Cambridge, England.

Weimann, G. (1983). The strength of weak conversational ties in the flow of information and influence. *Social Networks*, **5**: 245–267.

Wellman, B. (1981). Applying network analysis to the study of social support. *Social Networks and Social Support*, ed. B. H. Gottlief. Sage Publications, Beverly Hills, CA, pp. 171–200.

West, J. and Gallagher, S. (2006). Challenges of open innovation: The paradox of firm investment in open source software. *R&D Management*, **36**: 315–328.

Williams, K. Y. and O'Reilly, C. A. (1998). Forty years of diversity research: A review. *Research in Organizational Behavior*, eds. B. M. Staw and L. L. Cummings, Vol. 20. JAI Press, Greenwich, CT, pp. 22–140.

Withey, M., Daft, R. L. and Cooper, W. H. (1983). Measures of Perrow's work unit technology: An empirical assessment and a new scale. *Academy of Management Journal*, **26**: 45–63.

Zenger, T. R. and Lawrence, B. S. (1989). Organizational demography: The differential effects of age and tenure distributions on technical communications. *Academy of Management Journal*, **32**: 353–376.

Zobel, A. (2017). Benefiting from open innovation: A multidimensional model of absorptive capacity. *Journal of Product Innovation Management*, **34**: 269–288.

Zohar, D. and Luria, G. (2005). A multilevel model of safety climate: Cross-level relationships between organization and group-level climates. *Journal of Applied Psychology*, **90**: 616–628.

Appendix A. Endnotes

Note 1
Preliminary data analysis indicated that the number of links in an individual's technical advice network was highly correlated with the number of years that individual had been a member of the organization. Individuals who have been members of the organization for a number of years tended to list a greater number of names in their technical advice network than did more recent members. Thus, one reason to use proportions is to standardize the size of the technical advice network. Previous studies have operationalized multiplexity as a proportion or ratio [Ray and Miller (1991); Ibarra (1995)].

As the emphasis of this study is on the group's ability to find new or diverse information through its weak boundary-spanning technical advice links, and not the amount or quantity of this information, the *number* of boundary-spanning technical advice links is probably not as important as the *proportion* of the total number of technical advice links (both inside and outside the project group; strong, duplex, and weak) that are weak boundary-spanning links (links going outside the project group that do not overlap with a respondent's work or social networks). For the groups in this study, the proportion of weak boundary-spanning technical advice links (the dependent variable) ranged from a minimum of 0.0 to a maximum of 0.36 with 0.15 being the mean value.

Note 2
Since strength of tie is sometimes measured as frequency of interaction, a check was made to determine how operationalizing weak ties in terms of link multiplexity compared to measures based on interaction frequency. Therefore, survey respondents were asked to indicate how frequently they interacted with each of the individuals listed in the three networks described above. A *t*-test indicated that the average frequency of interaction for technical advice links coded as strong based on link multiplexity was statistically higher (at the 0.05 level of significance) when compared with the average frequency for technical advice links coded as weak. In addition, there is a correlation of 0.63 between strength of tie measured in terms of link multiplexity (i.e. the proportion of the technical advice network that go outside the project group and that do not overlap with either of the other two networks) and strength of tie measured in terms of interaction frequency (i.e. the proportion of the technical advice network where the frequency of interaction is less than once every two weeks). Other studies [e.g. Ray and Miller (1991)] have also found a high correlation between frequency of contact and link multiplexity.

The organization in which this research was conducted is organized into four hierarchical levels: branches (project groups), departments, directorates, and top management. The weak boundary-spanning technical advice links that are the focus of this study go to individuals outside of the project group but within the organization. A supplemental analysis of the technical advice network data was conducted to examine how many boundaries were crossed by these weak technical advice links (e.g. do these links go to people within the same department or outside the department?). It was found that weak technical advice links are more likely to go to individuals outside the same department while technical advice links that overlap with either the work network or social network (strong or duplex links) are more likely to go to people within the same department. Thus, by eliminating technical advice links that overlap with either of the other two networks (i.e. the work and social networks), the technical advice links that remain are more likely to go to individuals in more "distant" parts of the organization.

Note 3

The 30 top-level managers asked to rate the performance of each project group included department heads responsible for some project groups in the study as well as heads of departments that were not responsible for any of the project groups included in this study. Not all 30 managers were equally familiar with every project group so there was considerable variance in the ratings of these 30 managers. For example, department heads tended to rate project groups in their department higher than they rated groups in other departments. A few managers did not rate a couple of groups in "distant" parts of the organization, believing that they were not familiar enough with that group to provide an accurate assessment.

Several different approaches to analyzing the data were tried before settling on the approach used by Tushman [1979] which is to divide the project groups into high and low performers. For example, a regression was run with group performance ratings as the dependent variable and with the independent variable used in this study, but the regression was not statistically significant. A regression equation was also run that included all 49 project groups with a dummy variable to control for group performance. However, the results were not statistically significant.

Note 4

Although not the focus of this paper, a regression equation was run with strong, multiplex links as the dependent variable. Strong multiplex links contain all three components: work, technical advice, and social. The regression results for high-performing groups are similar in form to that for weak links shown in Table 5, indicating that the proportion of strong links is relatively high when the proportion of weak links is also relatively high and when the proportion of duplex links is relatively low. Thus, consistent with the arguments of Hansen [1999], when weak boundary-spanning links are used to bring into the group the information needed to solve difficult-to-analyze task-related problems, the proportion of strong links is relatively high. These strong links within the project group may help the group to integrate and use the diverse information provided by the boundary-spanning technical advice links [Tushman (1977); Mors (2010); Michelfelder and Kratzer (2013); Levin *et al.* (2016); Rost (2011); Tiwana (2008)]. This suggests that high-performing project groups may adjust their proportion of weak, duplex, and strong links to fit the degree of task uncertainty they face and the amount of information readily available from other members of their group.

A similar regression analysis was performed for low-performing groups but was not statistically significant. Perhaps, low-performing groups lack the strong links needed to effectively use and integrate the diverse knowledge that boundary spanners bring into the group by way of their weak links.

Appendix B

B.1. *Questionnaire items*

All items were measured using a Likert-type scale. Respondents indicated the extent to which they agreed or disagreed with each statement, responding on a 7-point scale where 1 = strongly disagree and 7 = strongly agree. Note that a low numerical score for task analyzability means that tasks are easy to analyze (i.e. analyzability is high) and a high numerical score means that tasks are difficult to analyze (i.e. analyzability is low).

Task analyzability:

I often encounter difficult problems in my work for which there are no immediate or apparent solutions.
I frequently encounter problems in my work which require substantially different methods or procedures for doing it.
There is an understandable sequence of steps that can be followed in doing my job. (reversed)*

Group heterogeneity:

Members of my work group vary widely in their skills and abilities.
My work group contains members with widely varying backgrounds.
Members of my work group vary widely in their amount of experience.

Tolerance of differences:

A person can do things differently around here without provoking a negative reaction.
In this organization, people are expected to confine their interest to their specialized skills.
Members of this organization feel free to express different opinions or ideas.
The best way to get along in this organization is to think the way the rest of the group does.
Around here, a person can get into a lot of trouble by being different.

*Although a three-question scale was originally planned for task analyzability, the alpha of this scale was 0.60. A factor analysis revealed that the factor loading for the last question was much lower than the first two questions. Since adding the third question reduced the alpha for the scale to 0.60, this last question was not used. Eliminating the last question increased the alpha to 0.70.

Biography

Donald O. Wilson earned his Ph.D. degree at the University of California, Irvine and has been at member of the management faculty in the Saunders College of Business at the Rochester Institute of Technology since 1987. He holds a BS degree in Electrical Engineering, Oklahoma State University, a MS degree in Systems, and a Master of Public Administration from the University of Southern California. Before his recent retirement from RIT, he served as Director of the Executive MBA program, the Director of Graduate Business Programs, and the Associate Dean at the Saunders College of Business. He has served as an advisor and consultant on implementing an Air Force strategic management and planning system and served as a consultant on implementing Total Quality Management within the U.S. Air Force. His research interests include strategic management, organizational design issues, and methods of changing communication patterns among scientists and engineers in order to enhance innovation and creativity. He has done extensive research into problems related to organizational management for the U.S. Navy and was previously under contract with the Navy to evaluate engineering fleet support activities — including job performance of engineers working on high-tech weapons systems.

John E. Ettlie is the Rosett Professor for Research in the Saunders College of Business, Rochester Institute of Technology. He earned his Ph.D. at Northwestern University and has held appointments since then at the University of Illinois Chicago, De Paul University, the Industrial Technology Institute, the University of Michigan, the U.S. Business School in Prague, Catolica University in Lisbon, Portugal, and Deusto University in Bilbao, Spain. Dr Ettlie has published over 125 articles and he has authored seven books, including the second edition of his textbook titled *Managing Innovation*. His current research interests include *social learning theory and entrepreneurship*, *cognitive style and creativity*, *new product development*, types of *open innovation*, *dynamic capabilities in R&D*, and *supply chain innovation*. He finished an NSF-sponsored study on R&D in the automotive industry and co-principal investigator of an NSF-sponsored workshop on globalization of engineering and R&D. He recently co-authored an article on open innovation in the Spanish Banking industry in the *Academy of Management Perspective*, Vol. 28, No. 1, February 2014, 76–91. Dr Ettlie was recently ranked 8th in the world among innovation management scholars (*Journal of Product Innovation Management*, 2012, 29/2). Dr Ettlie has raised $750,000 in research grant support since joining RIT in 1999. His most recent book with Dr Ron Hira, was published in May 2018 titled: *Engineering Globalization Reshoring and Nearshoring*: *Management and Policy Issues* (The World Scientific Press, ISBN 978-981-3147-02-7).